Algorithm Formula

Score formula is:

$$SC = \frac{(clicks \cdot w_c + votes \cdot w_v + comments \cdot w_m)}{(currentTime - postTime + 2)^G}$$

- ➔ Clicks, votes, comments get a certain weight (0.3, 0.4, 0.3)
- ➔ currentTime, postTime both in minutes
- ➔ +2 prevents division by zero and avoids spikes. If we remove +2, the score goes to infinity when the post is new.

Simpler:

$$score = \frac{E}{(t + C)^G}$$

- ➔ E is event
- ➔ t is time
- ➔ C is so called constant offset

Weights

We want to push posts which are engaged with:

1. **Number of votes**
2. **Number of comments**
3. **Number of clicks**

Votes: 0.6

Comments: 0.3

Clicks: 0.1

But:

Post X: 100 clicks, 20 votes, 5 comments => 23.5

Post Y: 300 clicks, 5 votes, 2 comments => 33.6

Which states that the clicks are more important than the votes. Big L. Not what we want.

So, to put everything to the same scale, normalization:

$$\text{normalized\_clicks} = \frac{clicks}{\text{avg\_clicks\_per\_post}}$$

$$\text{normalized\_votes} = \frac{\text{votes}}{\text{avg\_votes\_per\_post}}$$

$$\text{normalized\_comments} = \frac{\text{comments}}{\text{avg\_comments\_per\_post}}$$

So, the full finalized formula:

$$SC = \frac{(\text{normalized\_}clicks \cdot 0.6 + v\text{normalized\_}otes \cdot 0.3 + \text{normalized\_}comments \cdot 0.1)}{(currentTime - postTime + 2)^G}$$

Optional Comments Weights

$$Comment\ Score = base + b_l + b_r$$

e.g.

$b_l = length\_bonus\ =\ 0.6\ *\ ((length\_chars\ \%\ 100)\ /\ 100)$

$b_r = replies\_bonus = 0.12 \times min(numberReplies, 10)$

- ➔ 10 is br is the max_replies_cap